PVST+ and MSTP Interview Questions with Interview-Ready Answers

Q1. What is PVST+?

PVST+ (Per VLAN Spanning Tree Plus) is a Cisco-proprietary enhancement of STP. It creates a separate spanning-tree instance for every VLAN. This allows each VLAN to have its own root bridge and forwarding topology. It improves load balancing and traffic distribution across redundant links.

Q2. Why is STP required in switched networks?

STP prevents Layer 2 loops created by redundant links between switches. Without STP, broadcast storms and duplicate frames can consume network bandwidth. It also prevents MAC address table instability. STP provides redundancy while maintaining a loop-free topology.

Q3. What is a Root Bridge?

The Root Bridge is the central reference switch in a spanning-tree topology. All path calculations are performed relative to the root bridge. Every non-root switch determines its best path toward the root. A stable root bridge helps ensure predictable network behavior.

Q4. How is Root Bridge elected?

The election is based on the Bridge ID value. Bridge ID consists of bridge priority and MAC address. The switch with the lowest Bridge ID becomes the Root Bridge. If priorities are equal, the lowest MAC address wins.

Q5. Difference between STP, PVST+, and Rapid PVST+?

Traditional STP uses a single spanning-tree instance for all VLANs. PVST+ creates a separate instance for each VLAN. Rapid PVST+ provides the same per-VLAN operation but uses the faster convergence mechanisms of RSTP. Rapid PVST+ significantly reduces recovery time after failures.

Q6. What are Root Port, Designated Port, and Blocking Port?

A Root Port is the best path from a non-root switch to the root bridge. A Designated Port forwards traffic for a network segment. A Blocking Port is placed in a non-forwarding state to prevent loops. These port roles ensure a loop-free Layer 2 environment.

Q7. What is the default bridge priority?

The default bridge priority is 32768. This value is combined with the VLAN ID in PVST environments. Administrators can lower the priority to influence root bridge selection. Lower priority values are preferred during elections.

Q8. How does PVST+ provide load balancing?

PVST+ allows different VLANs to have different root bridges. Network administrators can control which links are preferred for each VLAN. This prevents a single link from carrying all traffic. As a result, redundant links are utilized more efficiently.

Q9. What is MSTP?

MSTP stands for Multiple Spanning Tree Protocol based on IEEE 802.1s. It allows multiple VLANs to be mapped to a single spanning-tree instance. This reduces the number of STP calculations in the network. MSTP is commonly deployed in large enterprise environments.

Q10. Why is MST preferred over PVST in large networks?

PVST creates a separate STP instance for every VLAN, which increases overhead. MST reduces the number of active spanning-tree instances by grouping VLANs. This lowers CPU and memory consumption on switches. It also simplifies network management and scalability.

Q11. What is an MST Region?

An MST Region is a collection of switches sharing the same MST configuration. All switches in the region must have identical region parameters. The region appears as a single logical switch to other STP domains. This improves efficiency and consistency.

Q12. Which parameters must match to form the same MST Region?

The region name must be identical on all switches. The revision number must also match. VLAN-to-instance mappings must be exactly the same. Any mismatch causes switches to consider each other part of different regions.

Q13. What is the purpose of the Revision Number?

The revision number identifies a specific MST configuration version. It helps administrators track configuration changes within a region. All switches in the region must use the same revision number. It is one of the values used in MST consistency checks.

Q14. How are VLANs mapped to MST Instances?

VLANs are assigned to MST instances inside MST configuration mode. Multiple VLANs can share one MST instance. Each instance operates with its own spanning-tree calculation. This reduces the number of spanning-tree processes required.

Q15. What is the difference between MSTP and Rapid PVST+?

Rapid PVST+ runs a separate rapid spanning-tree instance for every VLAN. MSTP groups multiple VLANs into a smaller number of instances. MSTP is more scalable in environments with many VLANs. Both provide fast convergence, but MSTP requires fewer resources.

Q16. How does MST reduce CPU utilization?

Instead of maintaining hundreds of spanning-tree instances, MST uses only a few instances. Fewer topology calculations are required during network changes. This reduces processor and memory consumption. The result is better scalability and performance.

Q17. What happens if MST configurations do not match?

Switches with different MST parameters form separate MST regions. The connection between them becomes a boundary port. STP information is exchanged differently across the boundary. This can affect path selection and load-balancing behavior.

Q18. How is a Root Bridge selected in MST?

Root bridge election occurs separately for each MST instance. The switch with the lowest Bridge ID becomes the root for that instance. Administrators often configure specific switches as primary and secondary roots. This provides predictable traffic flow and redundancy.